

Cairo Geospatial Chatbot — Test Cases

Section 1: Query Tests (with expected numeric answers)

#	Question	Expected Answer	Tests	Select Layers
1	Find pharmacies within 500 meters of Cairo University.	6 pharmacies (closest: El Nasr 90m, El Horreya 110m, Al Safwa 240m, El Horreya 2 260m, El Rehab 469m, El Karma 489m)	Radius / ST_DWithin	pharmacies, universities
2	Show schools inside Nasr City.	16 schools	Contains / ST_Contains	schools, neighborhoods
3	Which parks intersect flood zones?	5 parks (El Mostakbal, El Salam, Al Andalus, Al Andalus 2, El Yasmin 2)	Intersects (poly × poly)	parks, flood_zones
4	Nearest hospital to Downtown School.	El Horreya Specialized Hospital, ~1101 m (1 row)	Nearest / KNN	hospitals, schools
5	Restaurants within 300 meters of metro stations.	69 restaurants	Radius	restaurants, metro_stations
6	Show all schools within Downtown.	11 schools	Contains	schools, neighborhoods
7	Hospitals within 1 km of the Nile.	13 hospitals	Buffer (line ref)	hospitals, rivers
8	Which roads cross the river?	7 roads (El Tahrir, El Nil, El Horreya, Ramses, El Ahram, Gamal Abdel Nasser Axis, Ring Road)	Intersects (line × line) / ST_Crosses	roads, rivers
9	Find parking near hospitals.	36 parking lots within 200 m (every hospital has one)	Nearest / radius	parking, hospitals
10	What is the closest pharmacy to City Mall?	El Gomhoria Pharmacy 6, ~98 m (1 row)	Nearest	pharmacies, shopping_malls

Section 2: Visualization / Routing Tests (qualitative — expected behavior)

#	Question	Expected Result
11	Display restaurant density.	Heatmap with a strong hotspot over Downtown (178 restaurants; next: Nasr City 28, Zamalek Isle 25)
12	Show all restaurants downtown.	~178 markers → marker clustering kicks in when zoomed out
13	Walk/drive from Cairo University to City Mall.	A route polyline + start/end markers + distance & time (once Valhalla/OSRM is wired)
14	Show areas reachable within 15 minutes from Downtown.	An isochrone polygon containing many restaurants/hospitals/pharmacies (Valhalla)